



GE VERNOVA

ADMS

16.25.1

Relational Database Installation Guide

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Change History

Date	Change Description
Jun 7, 2023	ADMS 3.16: • Editorial and formatting updates.
Aug 3, 2023	ADMS 3.16.0.2: • Added Enabling MSDTC (Microsoft Distributed Transaction Coordinator). • Updated Upgrading in SQL Server.
Oct 20, 2023	ADMS 16.1.0: • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
Dec 5, 2023	ADMS 16.3.0: • Updated RDBMS Support.
Apr 1, 2024	ADMS 16.7.0: • Updated document format and styling.

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1. Database Overview and Installation Prerequisites

This chapter provides an overview of the ADMS relational databases, and describes the conditions that must be met prior to installing the databases.

1.1. Database Overview

1.1.1. Database Content

ADMS contains two relational databases. The Archive database contains historical data from the Distribution Management System, Outage Management System, Switching Order, and Safety Document applications for reporting purposes. The DNOM database contains Distribution Network Object Model (DNOM) data.

The ADMS Archive database is loaded by the ADMS Archive application and the Customer Loader tool. The database stores Distribution Power Flow (DPF) solutions (summary data only), "advanced application" (FISR, LVM, AFR) solutions, incidents, crews, calls, meter events, meter statuses, switching orders, safety documents, customer data, and reference data. Some company-specific static data (for example, cities, service centers, etc.) are loaded manually.

The ADMS DNOM database is loaded by the ADMS DNOM Database Loader application. It stores DNOM objects such as stations, feeders, switches, and measurements. This DNOM database is used by the Archive application to supplement the device data provided by archived transactions, particularly safety documents created from SCADA displays that do not have DNOM device data, so that the archived device data available for reporting is complete. It may also be used for other reporting or external interfaces that require DNOM data, or to facilitate manual access to network model data that is not easily accessible from binary station model (*.mod) files.

The Archive application will run successfully even if the DNOM database does not exist or is not loaded. In this case, DNOM data is not available to supplement device data on archived transactions. However, transactions are still archived with the data available on the transaction records. An Archive application configuration parameter (UseDnomDb) controls whether the Archive application attempts to look up data from the DNOM database. If the DNOM database does not exist or is not loaded, UseDnomDb should be set to "false".

Note that this relational DNOM database is separate from the memory-resident DNOM database used by the DMS server. Both of these DNOM databases are loaded from the same station model files. But since the DNOM databases are completely separate, queries against the relational DNOM database will not impact the performance of the memory-resident DNOM database used by the DMS server.

1.1.2. RDBMS Support

The Archive database may be on any of the following:

- Oracle Database 19c (certified)
- Oracle Database 12cR2 (supported)

On Oracle Database 12c, the Archive database may be a pluggable database (PDB).

- Oracle Database 11gR2 (supported)
- Microsoft SQL Server 2022 RTM (certified)
- Microsoft SQL Server 2019 RTM (certified)
- Microsoft SQL 2019 RTM (certified)
- Microsoft SQL Server 2014 SP3 CU4 (supported)
- Microsoft SQL Server 2012 SP4 (supported)

The DNOM database may be on any of the following:

- Oracle Database 19c (certified)
- Oracle Database 12cR2 (supported)
- Oracle Database 11gR2 (supported)
- Microsoft SQL Server 2022 RTM (certified)
- Microsoft SQL Server 2019 RTM (certified)
- Microsoft SQL Server 2014 SP3 CU4 (supported)
- Microsoft SQL Server 2012 SP4 (supported)

1.1.3. Database Deliverables

The ADMS relational database software is delivered as a Windows installer named `ADMS_relational_database.msi`. This installation package includes all the necessary files to create and upgrade the ADMS Archive and DNOM databases in an existing Oracle or SQL Server instance. Refer to documentation provided by Oracle or Microsoft for instructions to create the database server instance in which these databases reside.

1.1.4. SQL Server Logins

In SQL Server, the DNOM database includes a schema named "DNOM" owned by user dbo, and the Archive database includes a schema named "ARCHIVE" owned by user dbo. In order for the DNOM Database Loader and Archive applications to connect to the DNOM and Archive databases, you must create logins for the applications, and associate those logins with a user (for example, dbo) and default schema (for example, dbo) in each database. For more information, see [How to: Create a SQL Server Login^{\[1\]}](#).

If you want the DNOM Database Loader and Archive applications to connect to the DNOM and Archive databases via a login with SQL Server authentication instead of Windows authentication, you must create users in each database and associate the logins with those users. You may create users specifically for the applications, or you may grant permission for other users that are associated with logins to impersonate the application user. The instructions to create DNOM and Archive databases in SQL Server include an optional step to create users specifically for the applications.

GE recommends that you create SQL Server logins that use Windows authentication rather than SQL Server authentication. This enables integrated security, so that database connection strings specified in configuration files (for example, `ArchiveAppConfig.xml`) do not need to include a password. Regardless of whether you use Windows authentication or SQL Server authentication, database connection strings specified in configuration files may be encrypted. For information about encrypting database connection strings, refer to the *ADMS Archive User Guide*.

1.1.5. SQL Server Data-Tier Application Packages

ADMS databases are deployed and upgraded on Microsoft SQL Server via Data-tier Application (DAC) packages. A DAC package is an XML file containing a manifest that defines all database engine objects used by the application. A DAC is deployed to an SQL Server database via a command script or via the Deploy Data-Tier Application wizard in SQL Server Management Studio. For more information, refer to [Understanding Data-tier Applications^{\[2\]}](#).

1.1.6. Historian Database Integration

The ADMS Archive database does not store SCADA data or ADMS tags. SCADA data and tags are archived by the Historian application and stored in the Historian database. The ADMS Archive database does not reference the Historian database, and the Historian database does not reference the ADMS Archive database.

To facilitate reporting, it is useful to allow tags to be accessed from a database user account that also has access to the ADMS Archive database. This can be done by granting SELECT object privileges on the TAGS table to a user, who may then access the TAGS table via the TAGS public synonym.

For example, to enable a user to access the TAGS table, execute the following SQL statement from a Historian user connection:

```
GRANT SELECT ON TAGS TO _username_;
```

Similar statements may be executed to grant access to other tables in the Historian database (for example, TAGS_KEY, if applicable). If the Historian database is in a separate instance than the user that needs access to the TAGS table, a database link is also required.

1.1.7. ADO.NET Data Providers

The ADMS Archive application accesses the Archive database through [ADO.NET Data Providers](#)^[3]. For an Oracle database, the supported data provider is [Oracle Data Provider for .NET \(ODP.NET\)](#)^[4]. (Do not use Microsoft's deprecated .NET Framework Data Provider for Oracle.) For a SQL Server database, the supported data provider is [.NET Framework Data Provider for SQL Server \(SqlClient\)](#)^[5]. Since the data provider is installed on the application server, not on the database server, installation instructions for the data provider are in the *ADMS Installation Guide*.

1.2. Installation Prerequisites

1.2.1. Installation Overview

The installation process for the ADMS relational database consists of creating a database schema in an existing database instance.

- For the Oracle Database, installation is done by executing a script.
- For the SQL Server, installation is done using the SQL Server Management Studio wizard.

Although the ADMS relational databases may be installed in separate instances, it is recommended that they be installed in the same instance. This document assumes that both databases are installed in the same instance.

The DNOM database contains only a few objects that are created at installation time. On Oracle, these include a package containing functions and a public synonym for the package. On SQL Server, these include functions (SQL Server does not support packages as Oracle does). Tables that contain the DNOM objects are created dynamically by DNOM Database Loader based on the DNOM objects that appear in station model files. Therefore, after creating the DNOM database but before running DNOM Database Loader, the database will not contain tables.

1.2.2. Preparing for Installation

This section explains the necessary requirements for a successful installation of the ADMS relational databases.

An RDBMS must be installed and available. For more information about how to install, set up, and configure a database instance, refer to the RDBMS vendor's documentation.

The ADMS relational database may be installed on any hardware supported by the RDBMS vendor.

The ADMS relational database may be installed on a basic installation of the RDBMS, without the need for many advanced features. However, this does not preclude the use of advanced features such as partitioning, replication, etc.

This guide provides some size recommendations for some specific parameters, but it is up to each

installation to determine their appropriate database configuration according to the volume of expected data.

1.2.2.1. SQL Server Network Configuration

To configure SQL Server to allow connections by the Archive application via a specific TCP port:

1. Open the SQL Server Configuration Manager.
2. Under SQL Server Network Configuration, select Protocols for <instanceName> (for example, Protocols for SQLEXPRESS).
3. Double-click on TCP/IP to open the TCP/IP Properties dialog box.
4. Select the IP Addresses tab.
5. In the IPAll section, set TCP Port to the desired port (for example, 1434).

This port number must be specified in the DbConnectionString configuration parameter in the `ArchiveAppConfig.xml` file so that the Archive application connects to the database via that port.

6. If necessary, also configure your firewall to enable connections via that port. Firewall configuration instructions are not provided in this document.

1.2.3. System Requirements

The software requirements are categorized as operating system requirements and database requirements.

1.2.3.1. Operating System Requirements

As the ADMS relational database is in an RDBMS, the supported operating systems are those supported by the RDBMS.

1.2.3.2. Database Requirements

The following are the database requirements for the installation of the ADMS relational database:

- The current release of ADMS supports archive databases on both Oracle and SQL Server. The minimum version of Oracle Database is 11gR2. The minimum version of SQL Server is SQL Server 2012.
- The Oracle database installation scripts support two tablespaces for each database, one for tables and another one for indexes. The default tablespace names for the Archive database are ARCHIVE_DATA and ARCHIVE_INDEX. The default tablespace names for the DNOM database are DNOM_DATA and DNOM_INDEX. If your database will use only one tablespace, enter that tablespace name at both the Data Tablespace Name and Index Tablespace Name prompts.

1.2.3.3. Size Estimates

The Archive tablespace size estimates provided in this section assume that the Archive database will contain historical incident and switching order data, but not safety documents or DMS data. The estimates are based on six years of the following (average) data:

- Calls per day: 3,000
- Customers out per day: 37,000
- Incidents per day: 600
- Chronology entries per incident: 10
- Restoration steps per incident: 2
- Switching orders per day: 500
- Switching steps per order: 10

The estimates for the data and index tablespaces are:

- Data tablespace: 100 MB
- Index tablespace: 70 MB

These estimates include provisions for peak values (for example, hurricanes, big storms, etc.) during the six years for the aforementioned data items.

In Oracle, when archiving customer data, you might need to enlarge the data and index tablespaces in the Archive database. Use the following command to automatically enlarge a tablespace when additional tablespace size is needed:

```
ALTER DATABASE DATAFILE '<Full path to a tablespace file, ARCHIVE_DATA or ARCHIVE_INDEX>' AUTOEXTEND  
ON NEXT 100M;
```

For example:

```
ALTER DATABASE DATAFILE 'D:\Oracle19\server\oradata\ORCL\ARCHIVE_DATA.DBF' AUTOEXTEND ON NEXT 100M;
```

[1] <http://msdn.microsoft.com/en-us/library/aa337562%28v=sql.105%29.aspx>

[2] <http://msdn.microsoft.com/en-us/library/ee240739.aspx>

[3] <http://msdn.microsoft.com/en-us/data/dd363565.aspx>

[4] <http://www.oracle.com/technetwork/topics/dotnet/index-085163.html>

[5] <http://msdn.microsoft.com/en-us/library/a6cd7c08.aspx>

2. Database Installation and Configuration

This chapter describes the ADMS relational database installation process.

2.1. Summary of Installation Steps

The main steps of the ADMS relational database installation and configuration are as follows:

1. If appropriate, drop (delete) existing ADMS DNOM or Archive databases.
2. Create the ADMS DNOM or Archive databases. The DNOM database should be created first because a package in the Archive database references a package in the DNOM database.
3. Load company-specific data in the Archive database.

2.2. Dropping an Existing Database

If you have an existing test database from a previous version, and you do not want to preserve the data in that database, you may want to drop the database and create a new one rather than upgrade the existing database.

Caution

Dropping a database results in the loss of all data in the database, so this should only be done on test databases that contain data you do not need to keep.

To drop an Oracle database:

1. Connect as a user with DBA privileges, rather than the user that owns the database to be dropped.
2. Drop the user that owns the database schema by using the CASCADE option.

The CASCADE option causes all objects owned by the user (that is, the database schema) to be dropped along with the user. The example below shows how to drop database user "archive_test" and all objects owned by that user. This example uses a SQL*Plus command, but a database user can also be dropped using a visual tool such as Toad for Oracle or Oracle Enterprise Manager.

```

Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\>sqlplus sys@ids as sysdba

SQL*Plus: Release 11.2.0.2.0 Production on Wed Oct 5 12:34:49 2011
Copyright (c) 1982, 2010, Oracle. All rights reserved.

Enter password:

Connected to:
Oracle Database 11g Enterprise Edition Release 11.2.0.2.0 - 64bit Production
With the Partitioning, OLAP, Data Mining and Real Application Testing options

SQL> drop user archive_test cascade;

User dropped.

SQL> quit
Disconnected from Oracle Database 11g Enterprise Edition Release 11.2.0.2.0 - 64
With the Partitioning, OLAP, Data Mining and Real Application Testing options
C:\>_

```

3. Create and populate a new database, as explained in the following sections.

To drop a SQL Server database:

- Right-click on the database in SQL Server Management Studio and select Delete from the context menu.

Note

The "Delete" context menu item in SQL Server Management Studio "drops" a SQL Server database. You may also drop a SQL Server database by executing a Transact-SQL "DROP DATABASE" statement.

2.3. Installing the DNOM Database

The following sections provide instructions for installing the ADMS DNOM database.

2.3.1. Installing in Oracle

To install the DNOM database schema into an Oracle instance:

1. Open `ADMS_relational_database.msi` and run the installer. The installation directory (for example, `D:\Eterra\Distribution\Database`) contains a `DNOM\Oracle` subdirectory, hereafter referred to as *dnomdb_install_dir*.
2. Open a command prompt and change directory to *dnomdb_install_dir*.
3. Using SQL*Plus, connect as a user with DBA privileges (for example, SYS or SYSTEM).
4. Execute the file `CreateDnomUser.sql` by entering the following command:

```
@CreateDnomUser
```

This script creates the DNOM user and schema (one package and one synonym).

5. Answer the following prompts (some of the prompts provide default values, which are used if no value is entered).
 - Enter the ADMS DNOM user name (default DNOM):
 - Enter the ADMS DNOM password (default DNOM):
 - Enter the ADMS DNOM database name (default=ADMS):
 - Enter the ADMS DNOM data tablespace name (default DNOM _DATA):
 - Enter the ADMS DNOM index tablespace name (default DNOM _INDEX):
 - Enter the temporary tablespace name (default=TEMP):
6. Check the following log file for errors: `CreateDnomSchema.log`

If errors are present, contact your database administrator or GE for assistance.

2.3.2. Installing in SQL Server

To install the DNOM database into a SQL Server instance:

1. Open `ADMS_relational_database.msi` and run the installer. The installation directory (for example, `D:\Eterra\Distribution\Database`) contains a `DNOM\SQL Server` subdirectory, hereafter referred to as *dnomdb_install_dir*.
2. In SQL Server Management Studio, on the Object Explorer window, open the Management folder. Right-click on Data-tier Applications and select "Deploy Data-tier Application" from the context menu.
3. On the Introduction page of the Deploy Data-tier Application wizard, click Next.
4. On the Select Package page, enter the path of the `DNOM.dacpac` file in *dnomdb_install_dir*, and then click Next.
5. On the Update Configuration page, change the default database name if needed, and then click Next.
6. On the Summary page, click Next to begin the deployment.
7. On the Deploy DAC page, wait for the deployment to complete. If all deployment steps complete successfully, click Finish. Otherwise, resolve the errors and repeat this process.

Note that the deployment attempts to create a SQL Server login (for example, `ADMSARC1\USER`) that is associated with the DNOM database user on the SQL Server instance from which `DNOM.dacpac` was extracted. This causes warnings that may be ignored.

8. When the wizard is finished, review the DNOM database in the Databases folder of SQL Server

Management Studio Object Explorer. In particular, verify that the database contains scalar-valued functions (in the Programmability\Functions\Scalar-valued Functions folder).

9. Create a SQL Server login for the DNOM Database Loader application. See [SQL Server Logins](#).
10. (This step is only necessary if you want to create a login and user specifically for the DNOM Database Loader application. See [SQL Server Logins](#).) Execute the PostDeployment.sql script to create a DNOM user for the ADMS DNOM Database Loader application. In SQL Server Management Studio, on the File menu, select Open, and then click File, or click the Open File toolbar button. Select the PostDeployment.sql file and click Open. If necessary, change the login name to the name of the login created in the previous step. Select the DNOM database from the Available Databases list, and then click Execute.

2.4. Installing the Archive Database

The following sections provide instructions for installing the ADMS Archive database.

2.4.1. Installing in Oracle

To install the Archive database into an Oracle instance:

1. If not done previously to install the DNOM database, open ADMS_relational_database.msi and run the installer. The installation directory (for example, D:\Eterra\Distribution\Database) contains an Archive\Oracle subdirectory, hereafter referred to as *arcdb_install_dir*.
2. Open a command prompt and change directory to *arcdb_install_dir*.
3. Using SQL*Plus, connect to a user with DBA privileges (for example, SYS or SYSTEM).
4. To create an Archive user and schema, enter the following SQL*Plus command:

```
@CreateArchiveUser
```

5. Answer the following prompts (some of the prompts provide default values, which are used if no value is entered).
 - Enter the ADMS Archive user name
(default=ARCHIVE):

Important

For Oracle 12c, with the Create as Container Database installation option selected, user and role names must start with C## or c## (for example, C##ARCHIVE).

- Enter the ADMS Archive password
(default=ARCHIVE):
- Enter the ADMS Archive database name
(default=ADMS):

- Enter the ADMS Archive data tablespace name (default=ARCHIVE_DATA):
 - Enter the ADMS Archive index tablespace name (default=ARCHIVE_INDEX):
 - Enter the temporary tablespace name (default=TEMP):
6. Check the following log file for errors: `CreateArchiveSchema.log`
- If errors are present, contact your database administrator or GE for assistance.

2.4.1.1. Verifying the Archive Database Installation

To verify that the ADMS Archive Database has been installed correctly:

1. Open a command prompt and change directory to *arcdb_install_dir*.
2. Using SQL*Plus (or another tool such as TOAD), connect as the user created for the ADMS Archive Database.
3. Enter the following command:

```
@VerifyInstallation
```

The output of running this script should be "SUCCESS".

4. If the output is not "SUCCESS", check the messages and contact your database administrator.

2.4.2. Installing in SQL Server

To install the Archive database into a SQL Server instance:

1. If not done previously to install the DNOM database, open `ADMS_relational_database.msi` and run the installer. The installation directory (for example, `D:\Eterra\Distribution\Database`) contains an `Archive\SQL Server` subdirectory, hereafter referred to as *arcdb_install_dir*.
2. Deploy the ADMS Archive Database data-tier application. On SQL Server with SQL Server Data Tools, this is done simply by executing the command script `PublishDac.cmd` (edit the database connection string in the script if necessary). On SQL Server without SQL Server Data Tools, use the "Deploy Data-tier Application" wizard.

To start the "Deploy Data-tier Application" wizard in SQL Server Management Studio, right-click on Data-tier Applications in the Management folder of the Object Explorer window and select "Deploy Data-tier Application". Then follow the steps below.

- a. On the Introduction page of the Deploy Data-tier Application wizard, click Next.
- b. On the Select Package page, enter the path of `ARCHIVE.dacpac` file for SQL Server in the relevant subdirectory in *arcdb_install_dir*, and then click Next.

- c. On the Update Configuration page, change the default database name if needed, and then click Next.
- d. On the Summary page, click Next to begin the deployment.
- e. On the Deploy DAC page, wait for the deployment to complete. If all deployment steps complete successfully, click Finish. Otherwise, resolve the errors and repeat this process.

Note that the data-tier application deployment attempts to create a SQL Server login (for example, ADMSARC1\USER) that is associated with the ARCHIVE database user on the SQL Server instance from which ARCHIVE.dacpac was extracted. This causes warnings that may be ignored.

3. When the DAC has been deployed (published), review the ARCHIVE database in the Databases folder of SQL Server Management Studio Object Explorer. In particular, verify that the database contains tables.

Select the database folder in SQL Server Management Studio Object Explorer, then right-click it and select Refresh. The Archive DB should appear.

4. Create a SQL Server login for the Archive application. See [SQL Server Logins](#).
5. (This step is only necessary if you want to create a login and user specifically for the Archive application. See [SQL Server Logins](#).) Execute the script `PostDeployment.sql` to create an ARCHIVE user for the ADMS Archive application.
 - a. In SQL Server Management Studio, on the File menu, select Open, and then click File, or click the Open File toolbar button.
 - b. Select the `PostDeployment.sql` file and click Open.
 - c. If necessary, change the login name to the name of the login created in the previous step.
 - d. Select the ARCHIVE database from the Available Databases list, and then click Execute.
6. Load static system data into the ARCHIVE database. This may be done either by executing the `LoadSystemData.cmd` file at a command prompt, or by executing the `LoadSystemData.sql` Transact-SQL script in SQL Server Management Studio. Select the ARCHIVE database from the Available Databases list, and then click Execute.

The `LoadSystemData.cmd` file may need to be edited to specify the correct server name, instance name, and database name for your ARCHIVE database. Similarly, the `LoadSystemData.sql` script may need to be edited to specify the correct path to the other SQL scripts that it calls. Note that company-specific data must still be loaded separately as described in [Archive Database Loading](#).

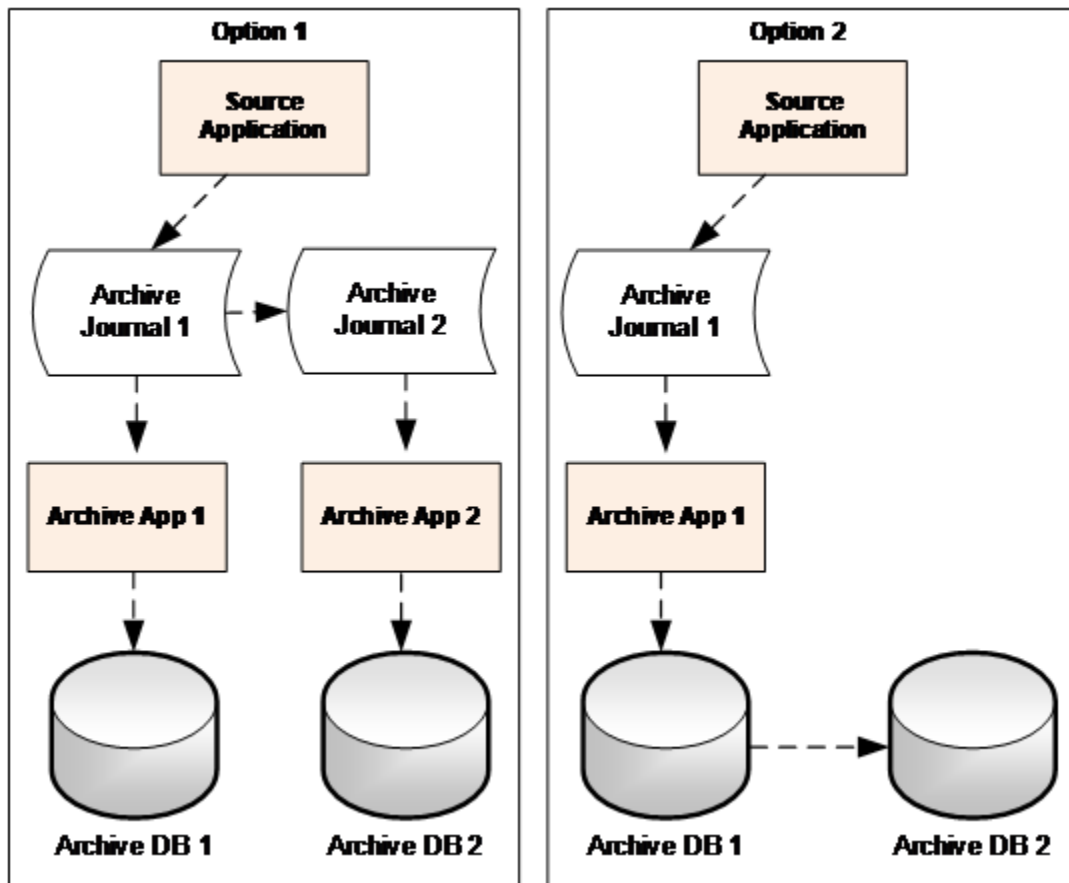
2.4.3. Enabling MSDTC (Microsoft Distributed Transaction Coordinator)

To enable MSDTC on the machine where Archive application is running and the SQL server in which Archive application is connected to:

1. Open Component Services from Start or Run dcomcnfg.
2. Navigate to Component Services > Computers > My Computer > Distributed Transaction Coordinator, right-click Local DTC, and click Properties.
3. On the Security tab, select the following:
 - Network DTC Access
 - Allow Remote Clients
 - Allow Inbound
 - Allow Outbound
 - No Authentication Required
4. Click OK to save your changes.
5. If prompted to restart the MSDTC service, click Yes. After the MSDTC service is restarted, close the Component Services window.

2.5. High Availability Configurations

Two high availability configuration options are supported.



Option 1

Two instances of the archive application run simultaneously and independently retrieve data from copies of the source data files (archive journals, safety documents, and reference data). The source data files must be replicated from the primary system to the standby system, using ADMS Net Data Replicator. (Since the archive application moves source files to an "Archived" folder when it is done with them, the two archive application instances must have separate copies of the source data files.) Each instance of the archive application updates a separate database.

This configuration does not use database replication software. Any manual database updates (for example, inserting a new feeder into the FEEDER table) must be made on both databases.

Option 2

A single instance of the archive application updates a single database. Database replication is used to create a second standby database.

The implementing project can choose which option to use. In either case, the archive application(s) should be started by the source application process starter. If the application's primary server fails, the assumption is that the source application has a failover mechanism for starting processes on a standby server. The archive application should be part of this failover mechanism.

3. Database Loading

This chapter discusses specific data loading tasks that are required after an ADMS relational database installation.

3.1. DNOM Database Loading

The ADMS DNOM database requires no manual data loading. Upon installation, the DNOM database contains no tables. Tables are created dynamically by the DNOM Database Loader.

3.2. Archive Database Loading

Most of the data in the ADMS Archive database is automatically loaded by the Archive application. However, there are a few tables in which the data needs to be loaded and maintained manually. These tables contain company-specific data that cannot be provided by GE. This data may be manually loaded using any database tool, such as Oracle SQL Developer or SQL Server Management Studio.

For the system to function correctly, or to enable certain types of reports to be produced, the following tables or columns are to be loaded manually (not by ADMS Archive):

- PD_AREA
- PD_OPERATION
- ORG_HIERARCHY_TYPE
- ORG_UNIT_TYPE
- ORG_UNIT_HIERARCHY (populate using the `ORG_UNIT_HIERARCHY.sql` script generated by the Customer List Compiler tool)
- ORG_UNIT (populate using the `ORG_UNIT.sql` script generated by the Customer List Compiler tool)
- SERVICE_CENTER (populate using the `SERVICE_CENTER.sql` script generated by the Customer List Compiler tool)
- SERVICE_CENTER_ORG (populate using the `SERVICE_CENTER_ORG.sql` script generated by the Customer List Compiler tool)
- STATION
- FEEDER
- MAJOR_EVENT
- Incident.Event_Id (foreign key to Major_Event.Event_Id)
- CUSTOMER (populate using the Customer Loader application, as described in the *Customer List Compiler User Guide*)

The following tables need to be populated to enable the Major Event Day (MED) functionality:

- MED_PARAMETERS
- MED_DAILY_SAIDI
- MED_SERV_CENTER_BMEE
- MED_BMEE_CUSTOMER_COUNT
- MED_EXCLD_INCIDENT_CAUSE
- MED_EXCLD_OUTAGE_TYPE
- MED_EXCLD_DEVICE_TYPE
- MED_BMEE

For more information about the Major Event Day (MED) functionality, refer to the *ADMS Archive User Guide*.

The ADMS installation includes a Fantasy Island data folder that contains SQL scripts to load some of these tables with data from GE's Fantasy Island demo model. You may use these scripts as templates for your own. Note that these scripts are intended only to load data into an empty database. If the database is not empty, you may encounter the following errors when executing the scripts:

```
ORA-02266: unique/primary keys in table referenced by enabled foreign keys
ORA-00001: unique constraint (ARCHIVE.PK_SERVICE_CENTER) violated
ORA-02291: integrity constraint (ADMS_NRT.FK_SERVICE_CENTER_ORG_UNIT_TYP) violated - parent key not found
```

For more information regarding the meaning or function of the data to be loaded, see [Archive Data](#).

4. Database Maintenance

4.1. Database Administration

Maintaining the ADMS relational database is similar to other databases:

- The operating system and database management system should be patched (after confirming with GE that ADMS is supported on the new versions).

As part of maintenance agreements, GE tests ADMS with operating system and database management system patch releases and provides the results.

- Disk space should be monitored and managed.
- Regular backups of pertinent data should be performed.

These are standard system and database administration tasks that do not need to be detailed in this document. For more information about maintaining the database, refer to the RDBMS vendor's documentation.

Tasks particular to the ADMS relational databases are described in the following sections.

4.2. Upgrading the DNOM Database

In general, the DNOM database may not be upgraded, but must be dropped, recreated, and reloaded. However, the default configuration of the DNOM Database Loader loads only fields that are required by the archive database, thereby minimizing the need to reinstall and reload the DNOM database.

The *ADMS Installation Guide* or *ADMS Release Notes* for each release indicates whether a database upgrade is necessary for that release. In some cases, only the stored functions called by the Archive application must be upgraded. In this case, functions may be recreated either by executing the `CreateFunctions.cmd` file at a command prompt, or by executing the `CreateFunctions.sql` Transact-SQL script in SQL Server Management Studio. The `CreateFunctions.cmd` file may need to be edited to specify the correct server name, instance name, and database name for your DNOM database. Similarly, the `CreateFunctions.sql` script may need to be edited to specify the correct path to the other SQL scripts that it calls.

4.3. Upgrading the Archive Database

Note

When upgrading to ADMS version 3.4 or later, restoration step data conversion is required after upgrading the database schema. This data conversion does not apply to users who do not archive OMS data, which includes restoration steps.

4.3.1. Upgrade Overview

Some releases of ADMS require database schema changes. The Archive database can be upgraded by executing a SQL script (on Oracle) or deploying a data-tier application package (on SQL Server).

ADMS Archive database version numbers are consistent with ADMS version numbers. For example, Archive database version 16.x.y is compatible with ADMS 16.x.y.

Generally, in a release in which the major version (for example, 16) or the minor version (for example, x) is incremented, the product has undergone significant changes so a database upgrade is necessary. In a patch release, a database upgrade may not be necessary.

Database upgrades apply all necessary schema changes to the current version. All schema changes resulting from a database upgrade are logged to a log file. The log file should be carefully reviewed to ensure that no errors were encountered before proceeding.

In some cases, a database upgrade involves data changes as well as schema changes. As of October 2015, a new utility named `ArchiveDataConverter.exe` is provided for data conversions. This program updates data in both Oracle and SQL Server databases. `ArchiveDataConverter.exe` should be run after the schema upgrade.

4.3.2. Upgrading in Oracle

To upgrade the Archive database in an Oracle instance:

1. Stop the ADMS Archive application if it is running.
2. Open `ADMS_relational_database.msi` and run the installer. The installation directory (for example, `D:\Eterra\Distribution\Database`) contains an `Oracle\archive` subdirectory, hereafter referred to as *arcdb_install_dir*.
3. Open a command prompt and change directory to *arcdb_install_dir*.
4. Using SQL*Plus, connect to the ADMS Archive database as the user that owns the ADMS Archive database schema (for example, "ARCHIVE") and execute the `UpgradeArchiveSchema.sql` script to upgrade the Archive database schema:

```
sqlplus.exe ARCHIVE/ARCHIVE@ORCL @UpgradeArchiveSchema
```

`UpgradeArchiveSchema.sql` generates a new upgrade script file, `upgrade_to_X.Y.sql`. This new file includes all the necessary calls to the actual scripts (`upgrade_to__X.Y_.sql` files in the `Oracle/upgrade` folder) that are executed to perform incremental version updates based on the current version and the target version. This new script is executed automatically.

The upgrade script does not fail if schema changes have already been made. For example, if the upgrade script attempts to drop a column that does not exist, it does not fail. This enables the upgrade script to be executed more than once. However, anything unexpected (not enough disk space, missing privileges, missing tablespaces, etc.) causes the upgrade script to fail. Appropriate messages are recorded in the `upgrade_to_X.Y.log` file.

5. Review the `upgrade_to_X.Y.log` file for errors.
6. If there are no errors, run the ADMS Archive application installed with the ADMS server kit.

To upgrade the Archive database schema versions 3.9 and earlier for the Enterprise OMS functionality:

1. Using SQL*Plus, connect to the ADMS Archive database as the user who owns the ADMS Archive database schema (for example, "ARCHIVE").
2. Execute the `ArchiveDDL_WorkModel_Oracle.sql` script to upgrade the Archive database schema:

```
sqlplus.exe ARCHIVE/ARCHIVE@ORCL @ArchiveDDL_WorkModel_Oracle
```

Note

The Archive database schema versions 3.10 and later includes the Enterprise OMS functionality.

4.3.3. Upgrading in SQL Server

Important

When upgrading the Archive database in an SQL Server instance using a command-line tool, such as `SqlPackage.exe`, add the `/p:BlockOnPossibleDataLoss=false` parameter to the command to deploy `ARCHIVE.dacpac`. This setting defines that the upgrade can proceed despite possible data loss. The data loss occurs in deprecated tables. (When using the "Upgrade Data-tier Application" wizard in SQL Server Management Studio, there is no option to continue upgrade without enabling the "Proceed despite possible loss of changes" option on the "Review Upgrade Plan" page.)

To upgrade the Archive database in an SQL Server instance:

1. Stop the ADMS Archive application if it is running.
2. Open `ADMS_relational_database.msi` and run the installer. The installation directory (for example, `D:\Eterra\Distribution\Database`) contains an `SQL Server\archive` subdirectory, hereafter referred to as *arcdb_install_dir*.
3. Manually execute the upgrade scripts stored in the `SQL Server\upgrade` subdirectory within *arcdb_install_dir*.

Important

This must be done before deploying `ARCHIVE.dacpac` in the next step.

4. When upgrading the Archive database for versions 3.6 and earlier, deploy the ADMS Archive Database data-tier application, `ARCHIVE.dacpac`, for version 3.7. Skip this step when upgrading the Archive database for version 3.7.

On SQL Server with SQL Server Data Tools, this is done simply by executing the command script

PublishDac.cmd (edit the database connection string in the script if necessary). On SQL Server without SQL Server Data Tools, use the "Upgrade Data-tier Application" wizard.

To start the "Upgrade Data-tier Application" wizard in SQL Server Management Studio, right-click on Data-tier Applications in the Management folder of the Object Explorer window and select "Upgrade Data-tier Application". Alternatively, right-click on the Databases folder in the Object Explorer window and select "Upgrade Data-tier Application". Then follow the steps below.

- a. On the Introduction page of the Upgrade Data-tier Application wizard, click Next.
- b. On the Select Package page, enter the path of ARCHIVE.dacpac from version 3.7 for SQL Server in the relevant dacpac_idms_370 subdirectory in *arcdb_install_dir*, and then click Next.
- c. On the Detect Change page, review the changes, and then click Next.
- d. On the Options page, select the "Rollback on failure" option, and then click Next.
- e. On the Review Upgrade Plan page, review the upgrade plan, and then click Next.
- f. On the Summary page, click Next to begin the upgrade.
- g. On the Upgrade DAC page, wait for the upgrade to complete. If all upgrade steps complete successfully, click Finish. Otherwise, review and resolve the errors and repeat this process.

When the DAC has been upgraded (published), review the ARCHIVE database in the Databases folder of SQL Server Management Studio Object Explorer.

5. When upgrading the Archive database for versions 3.7 and earlier, manually execute the `upgrade_to_3.8.sql` script stored in the SQL Server\upgrade subdirectory in *arcdb_install_dir*.
6. When upgrading the Archive database for versions 3.7 and earlier, deploy the ADMS Archive Database data-tier application, ARCHIVE.dacpac, for version 3.8.5. For more information, see step 3 above.
7. When upgrading the Archive database for versions 3.8.5 and earlier, manually execute the `upgrade_to_3.8.6.sql` script.
8. When upgrading the Archive database for versions 3.8.6 and earlier, deploy the ADMS Archive Database data-tier application, ARCHIVE_NRT.dacpac for version 3.8.6. For more information, see step 3 above.
9. Manually execute the `upgrade_to_3.9.sql` script.
10. Deploy the ADMS Archive Database data-tier application, ARCHIVE.dacpac, for your ADMS version. For more information, see step 3 above.
11. Execute the script `PostDeployment.sql` to create an ARCHIVE user for the ADMS Archive application:
 - a. In SQL Server Management Studio, on the File menu, select Open, and then click File, or click the Open File toolbar button.
 - b. Select the `PostDeployment.sql` file and click Open.
 - c. If necessary, change the login name to the name of the login used by the Archive application.

- d. Select the ARCHIVE database from the Available Databases list, and then click Execute.
12. If there are no errors, run the ADMS Archive application installed with the ADMS server kit.

4.3.3.1. SQL Server Authentication

Windows authentication is recommended for the database connection string used by the Archive application to connect to the database. With Windows authentication, the `ArchiveAppConfig.xml` file need not include a database user name and password. However, if the database server is in a different domain than the application server, you must use SQL Server authentication. You may also choose to use SQL Server authentication for other reasons. If you use SQL Server authentication, note that after a database upgrade, the SQL Server login used by the Archive application must be remapped to the ARCHIVE database and reassigned the `db_datareader` and `db_datawriter` roles. This is because upgrading a data-tier application as described above results in the creation of a new ARCHIVE database.

5. Archive Data Dictionary

For more information about the Archive database schema, refer to the Archive Data Dictionary. The data dictionary is available in HTML format for viewing in a web browser. The data dictionary files are in the ArchiveDataModel\Data Dictionary subdirectory of the ADMS relational database installation directory (D:\Eterra\Distribution\Database\Archive\Data Dictionary).

To open the data dictionary, open the `index.html` file in a web browser (for example, D:\Eterra\Distribution\Database\Archive\Data Dictionary\ADMS\schemas\ARCHIVE\index.html).

Note

The data dictionary is automatically generated from an Oracle database, therefore the data types indicated are Oracle data types. However, all of the database objects exist in a SQL Server database as well as in an Oracle database.

6. Sample Reports

ADMS provides some sample distribution and outage management reports that use near real-time and historical data from the Archive database. Power companies may use these reports directly, or as templates for their own custom reports.

- The `ADMS_sample_reports.msi` kit includes sample distribution and outage management reports using historical data from the Archive database. These reports are implemented using Microsoft SQL Server Reporting Services (SSRS), which is a feature of MS SQL Server. The reports are accessible from any web browser.

For more information about the ADMS Reports application, refer to the following:

- *Reports Installation Guide*
- *Reports User Guide*
- The `ADMS_relational_database.msi` kit includes sample reports for near real-time data. These reports are implemented using the Microsoft Power BI application, which allows generating various types of reports from different data sources, such as Excel files, CSV files, or MS SQL Server databases. The reports can be viewed by different types of users or user groups from the network or the internet.

By default, the `NRT_Report.pbix` file with reports is located in the `D:\Eterra\Distribution\Database\Archive\<DatabaseType>\reports` folder.

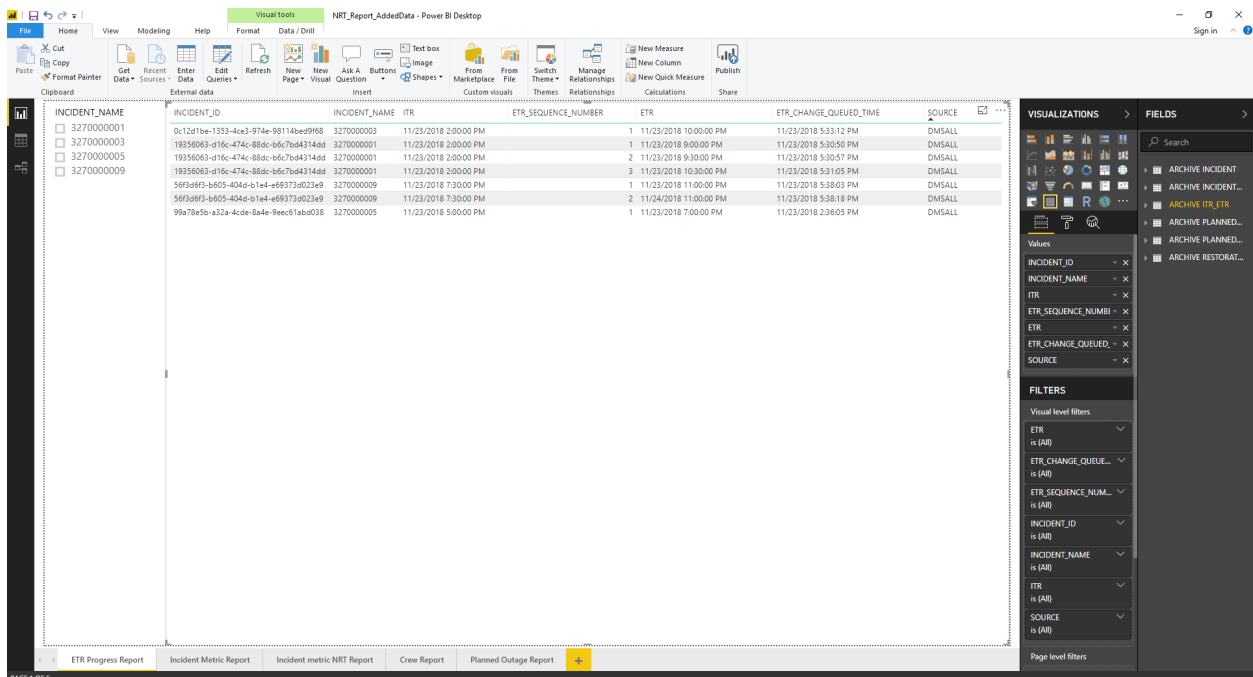


Figure 1. ETR Progress Report in Microsoft Power BI

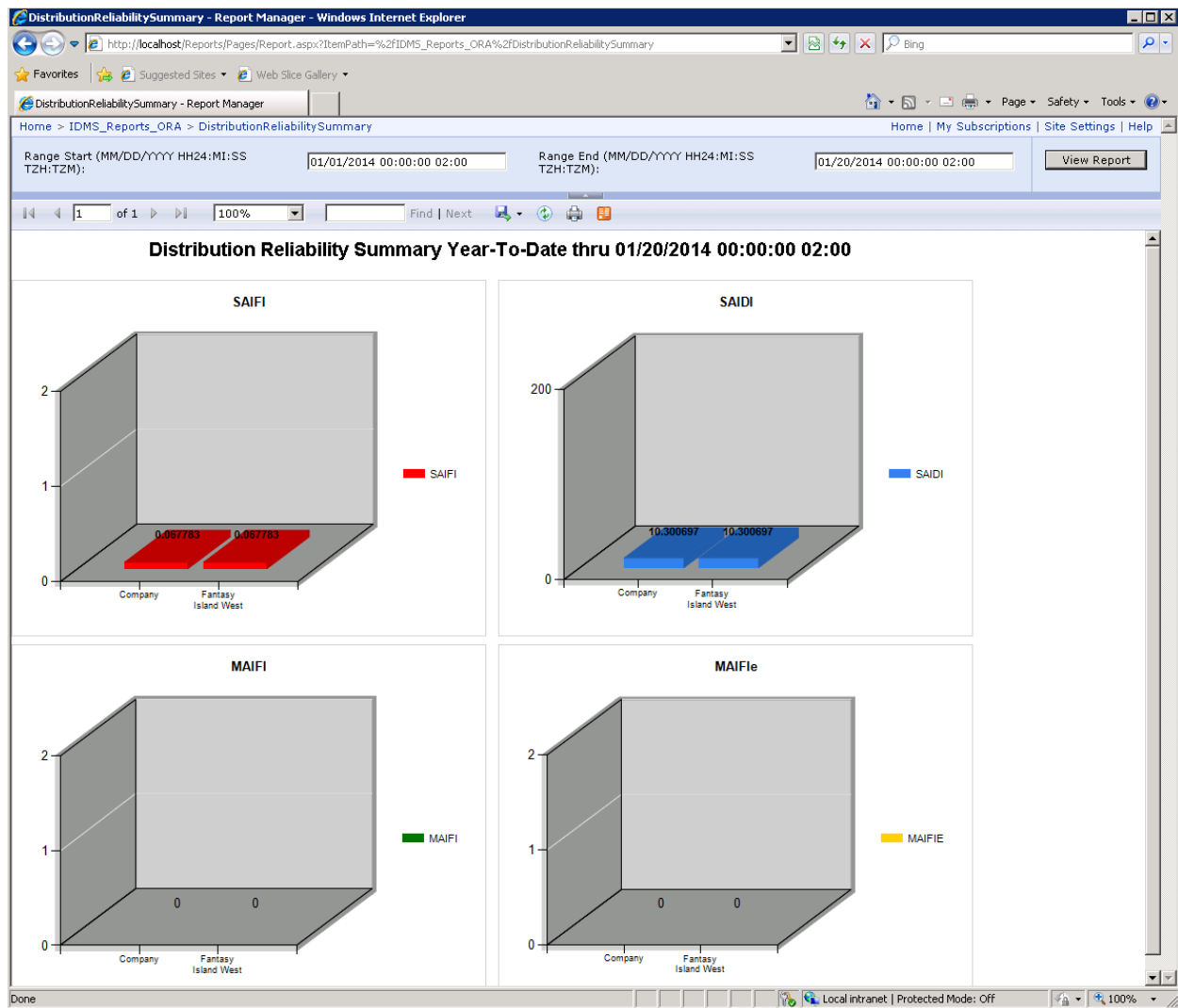


Figure 2. Distribution Reliability Summary Report in SSRS

To open a report in the Microsoft Power BI application:

1. Download and install Power BI Desktop from <https://powerbi.microsoft.com/en-us/desktop/>.
2. Open the Power BI Desktop application.
3. From the File menu, click Open and open a PBIX file.

Sample reports in NRT_Report.pbix include the following:

INCIDENT_NAME	INCIDENT_ID	INCIDENT_NAME	ETR	ETR_SEQUENCE_NUMBER	ETR	ETR_CHANGE_QUEUED_TIME	SOURCE
☐ 3270000001	0c12d1be-1353-4ce3-974e-98114bed9f68	3270000003	11/23/2018 2:00:00 PM	1	11/23/2018 10:00:00 PM	11/23/2018 5:33:12 PM	DMSALL
☐ 3270000003	19356063-d16c-474c-88dc-b6c7bd4314dd	3270000001	11/23/2018 2:00:00 PM	1	11/23/2018 9:00:00 PM	11/23/2018 5:30:50 PM	DMSALL
☐ 3270000005	19356063-d16c-474c-88dc-b6c7bd4314dd	3270000001	11/23/2018 2:00:00 PM	2	11/23/2018 9:30:00 PM	11/23/2018 5:30:57 PM	DMSALL
☐ 3270000009	19356063-d16c-474c-88dc-b6c7bd4314dd	3270000001	11/23/2018 2:00:00 PM	3	11/23/2018 10:30:00 PM	11/23/2018 5:31:05 PM	DMSALL
	56f3d6f3-b605-404d-b1e4-e69373d023e9	3270000009	11/23/2018 7:30:00 PM	1	11/23/2018 11:00:00 PM	11/23/2018 5:38:03 PM	DMSALL
	56f3d6f3-b605-404d-b1e4-e69373d023e9	3270000009	11/23/2018 7:30:00 PM	2	11/24/2018 11:00:00 PM	11/23/2018 5:38:18 PM	DMSALL
	99a78e5b-a32a-4cde-8a4e-9eec81abd038	3270000005	11/23/2018 5:00:00 PM	1	11/23/2018 7:00:00 PM	11/23/2018 2:36:05 PM	DMSALL

Figure 3. ETR Progress Report

INCIDENT_NAME	INCIDENT_ID	INCIDENT_NAME	INCIDENT_TYPE_CODE	PRIORITY	SERVICE_CENTER_ID	START_TIME	END_TIME	ETR	ITR
☐ 3270000001	19356063-d16c-474c-88dc-b6c7bd4314dd	3270000001	3	0.00	3LIONSSC	11/23/2018 11:49:49 AM			
☐ 3270000002	b3b1220e-369d-4010-8ce7-18ca224e96e9	3270000002	2	1.00	3LIONSSC	11/23/2018 11:50:25 AM			
☐ 3270000003	0c12d1be-1353-4ce3-974e-98114bed9f68	3270000003	0	1.00	3LIONSSC	11/23/2018 11:51:02 AM			
☐ 3270000004	6010aed3-93c2-4821-912a-53fd457f779	3270000004	7	0.00	3LIONSSC	11/23/2018 11:53:04 AM			
☐ 3270000005	99a78e5b-a32a-4cde-8a4e-9ec61abd038	3270000005	2	1.00	STHA	11/23/2018 2:34:53 PM			
☐ 3270000006	b80832ee-9ed3-496e-95ad-192bf1960814	3270000006	3	0.00	STHB	11/23/2018 5:13:59 PM	11/23/2018 5:17:35 PM	11/23/2018 7:30:00 PM	11/23/2018 7:30:00 PM
☐ 3270000007	a29d08df-34ef-4bd3-8ecf-cd52e163bf0f	3270000007	3	0.00	CTLB	11/23/2018 5:14:14 PM	11/23/2018 5:18:30 PM	11/23/2018 7:30:00 PM	11/23/2018 7:30:00 PM
☐ 3270000008	cd5f7752-64f0-4a3c-a22a-a9f3b1e833ca	3270000008	3	0.00	CTLB	11/23/2018 5:14:26 PM	11/23/2018 5:33:59 PM	11/23/2018 9:30:00 PM	11/23/2018 7:30:00 PM
☐ 3270000009	56f3d6f3-b605-404d-b1e4-e69373d023e9	3270000009	3	0.00	CTLC	11/23/2018 5:14:38 PM	11/23/2018 5:35:01 PM	11/24/2018 11:00:00 PM	11/23/2018 7:30:00 PM
☐ 3270000010	1a1d7b13-ca6a-435e-9c9c-81aedd0881ce	3270000010	0	1.00	CTLB	11/23/2018 5:40:08 PM			
☐ 3270000011	a956b445-574f-4f9b-b510-674445d1400c	3270000011	2	1.00	CTLB	11/23/2018 5:40:08 PM			
☐ 3270000012	6b790f58-da3b-4561-9cea-b068ad5896b6	3270000012	3	0.00	CTLC	11/23/2018 5:40:37 PM			
☐ 3270000013	23e664a2-56c4-42b4-92d3-a120a19b9124	3270000013	3	0.00	CTLC	11/23/2018 5:40:54 PM			
☐ 3270000014	3d54a7c9-f303-4407-bb97-20ea233abfd5	3270000014	3	0.00	CTLB	11/23/2018 5:41:32 PM			
☐ 3270000015	f7194ff7-0aab-4eb2-86cb-87ec6e09f372	3270000015	7	0.00	STHC	11/23/2018 5:46:34 PM			

Figure 4. Incident Metric Report

INCIDENT_NAME	INCIDENT_ID	INCIDENT_NAME	INCIDENT_TYPE_CODE	PRIORITY	SERVICE_CENTER_ID	START_TIME	ETR	ITR	FIRST_CALL_TIME	DEVICE_ID	PHASE
☐ 3270000001	19356063-d16c-474c-88dc-b6c7bd4314dd	3270000001	3	0.00	3LIONSSC	11/23/2018 11:49:49 AM				17674821	A
☐ 3270000002	b3b1220e-369d-4010-8ce7-18ca224e96e9	3270000002	2	1.00	3LIONSSC	11/23/2018 11:50:25 AM			11/23/2018 11:50:25 AM	17595438	A
☐ 3270000003	0c12d1be-1353-4ce3-974e-98114bed9f68	3270000003	0	1.00	3LIONSSC	11/23/2018 11:51:02 AM			11/23/2018 11:51:02 AM	SP_17595755	B
☐ 3270000004	6010aed3-93c2-4821-912a-53fd457f779	3270000004	7	0.00	3LIONSSC	11/23/2018 11:53:04 AM			11/23/2018 11:53:04 AM		None
☐ 3270000005	99a78e5b-a32a-4cde-8a4e-9ec61abd038	3270000005	2	1.00	STHA	11/23/2018 2:34:53 PM			11/23/2018 2:34:53 PM	14170970	A
☐ 3270000011	a956b445-574f-4f9b-b510-674445d1400c	3270000011	2	1.00	CTLB	11/23/2018 5:40:08 PM			11/23/2018 5:40:08 PM	156062670	ABC
☐ 3270000012	6b790f58-da3b-4561-9cea-b068ad5896b6	3270000012	3	0.00	CTLC	11/23/2018 5:40:37 PM				11641330	A
☐ 3270000013	23e664a2-56c4-42b4-92d3-a120a19b9124	3270000013	3	0.00	CTLC	11/23/2018 5:40:54 PM				11225003	C
☐ 3270000014	3d54a7c9-f303-4407-bb97-20ea233abfd5	3270000014	3	0.00	CTLB	11/23/2018 5:41:32 PM				8840342	A
☐ 3270000015	f7194ff7-0aab-4eb2-86cb-87ec6e09f372	3270000015	7	0.00	STHC	11/23/2018 5:46:34 PM					ABC

Figure 5. Incident Metric NRT Report

CREW_ID	CREW_ID	INCIDENT_ID	CREW_STATUS	INCIDENT_WORKING_STATUS	TIME_OF_THE_STATUS	CREW_WMS_LOGIN_ID	CREW_AVAILABLE_INDICATOR	USER_ID
☐ FI1502	FI1502	19356063-d16c-474c-88dc-b6c7bd4314dd	5	3	11/23/2018 12:00:46 PM	11101		1 DMSALL
☐ FI1520	FI1520	b3b1220e-369d-4010-8ce7-18ca224e96e9	5	3	11/23/2018 12:01:13 PM	11104		1 DMSALL
☐ FI1526	FI1526	0c12d1be-1353-4ce3-974e-98114bed9f68	5	3	11/23/2018 12:02:31 PM	11120		1 DMSALL
☐ FI1556	FI1526	0c12d1be-1353-4ce3-974e-98114bed9f68	7	5	11/23/2018 12:54:34 PM	11120		1 DMSALL
☐ FI1943	FI1556	0c12d1be-1353-4ce3-974e-98114bed9f68	5	3	11/23/2018 12:02:31 PM	11116		1 DMSALL
	FI1556	0c12d1be-1353-4ce3-974e-98114bed9f68	6	4	11/23/2018 12:54:19 PM	11116		1 DMSALL
	FI1943	0c12d1be-1353-4ce3-974e-98114bed9f68	5	3	11/23/2018 12:02:01 PM	5251		1 DMSALL
	FI1943	0c12d1be-1353-4ce3-974e-98114bed9f68	6	4	11/23/2018 12:02:21 PM	5251		1 Work Force Management
	FI1943	0c12d1be-1353-4ce3-974e-98114bed9f68	7	5	11/23/2018 12:02:51 PM	5251		1 Work Force Management
	FI1943	0c12d1be-1353-4ce3-974e-98114bed9f68	10	0	11/23/2018 12:05:21 PM	5251		1 Work Force Management

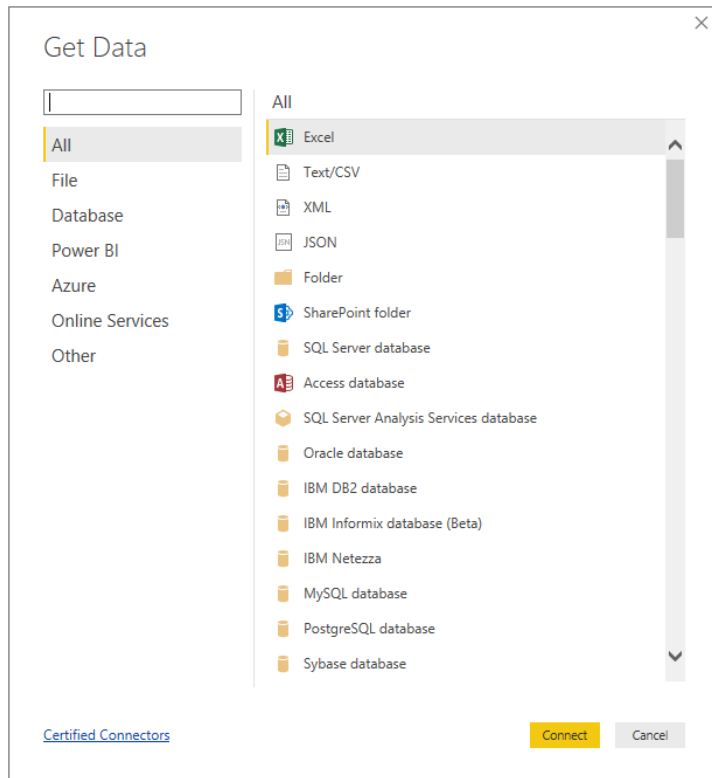
Figure 6. Crew Report

NAME	NAME	CREATED_TIME	PRIMARY_START_TIME	PRIMARY_END_TIME	ALTERNATE_START_TIME	ALTERNATE_END_TIME	USE_ALTERNATE_DATE	DEVICE_ID	REPRES_NAME	REPRES_NUMBER	OUTAGE_ME
☐ BAKER 50499	BAKER 50499	11/23/2018 5:27:22 PM	11/26/2018 12:00:00 AM	11/27/2018 12:05:00 AM	11/27/2018 12:00:00 AM	11/28/2018 12:05:00 AM	0	10115923	test	12345678	Dear Custom cause and th
☐ JUPITER 686501	SPYGLASS 223112	11/23/2018 5:52:37 PM	11/26/2018 12:00:00 AM	11/27/2018 11:05:00 PM	11/27/2018 11:00:00 AM	11/28/2018 7:05:00 PM	0	34811308	test5	123456789	Dear Custom cause and th
☐ PLUTO 503101	PLUTO 503101	11/23/2018 5:46:03 PM	11/28/2018 12:00:00 AM	11/29/2018 1:05:00 PM	11/29/2018 12:00:00 AM	11/30/2018 1:05:00 PM	0	19749233	test2	123456789	Dear Custom cause and th
☐ SPYGLASS 11...	SPYGLASS 11570	11/23/2018 5:47:37 PM	11/28/2018 12:00:00 AM	11/29/2018 11:05:00 PM	11/29/2018 12:00:00 AM	11/30/2018 11:05:00 PM	0	34811370	test3	12345678	Dear Custom cause and th
☐ SPYGLASS 22...	JUPITER 686501	11/23/2018 5:58:28 PM	12/5/2018 9:00:00 PM	12/9/2018 10:00:00 PM	12/13/2018 8:00:00 PM	12/22/2018 11:59:01 PM	1	19750180	test4	11234545467	Dear Custom and thank yo

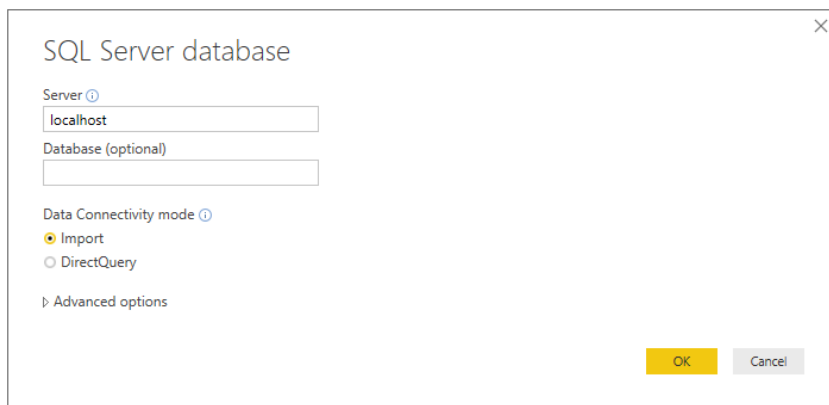
Figure 7. Planned Outage Report

To create a report using the Microsoft Power BI application:

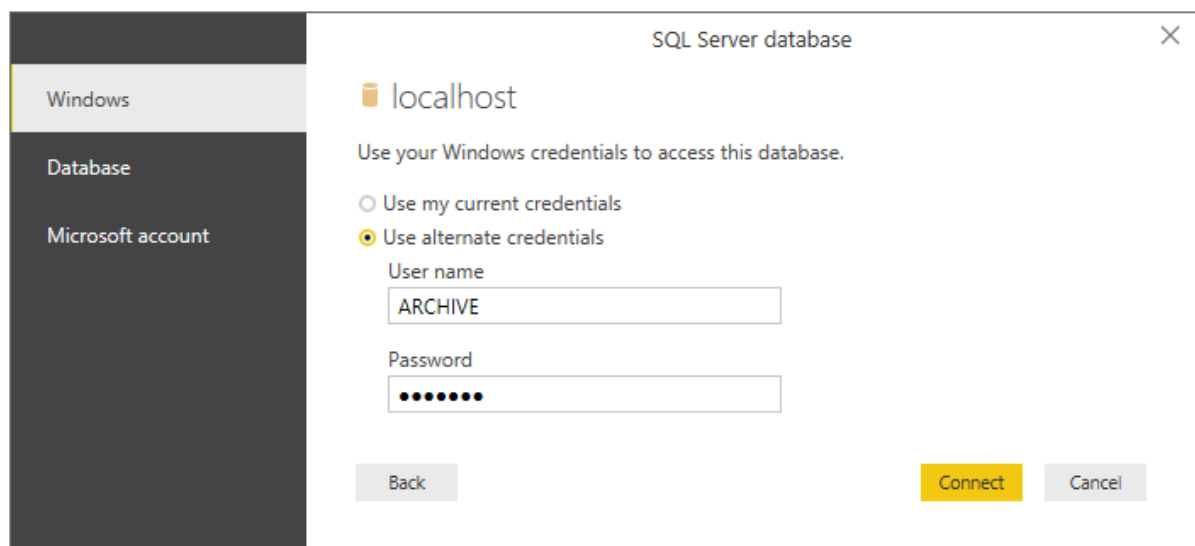
1. Open Power BI Desktop.
2. Click Get Data.
3. Select the database type that you use for storing archive data.



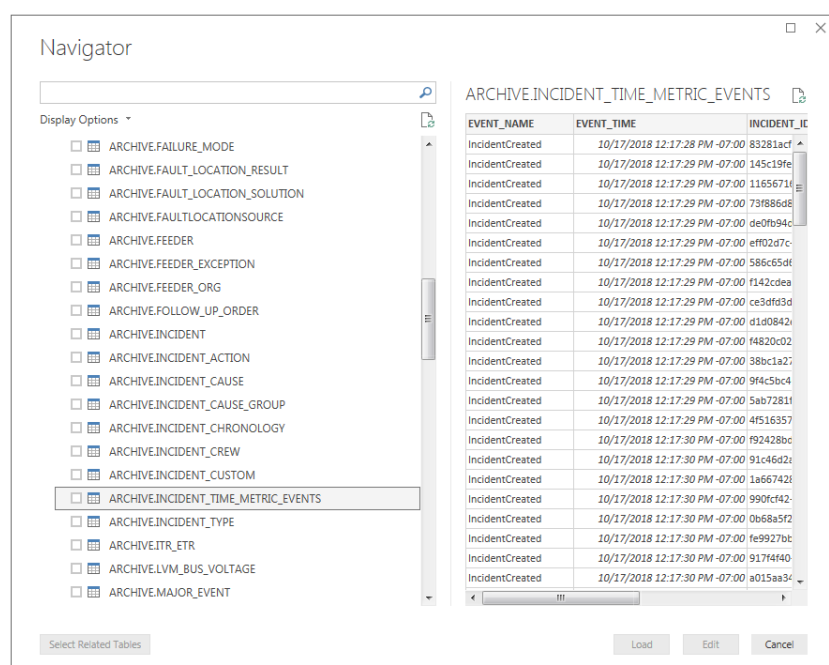
4. In the Server field, specify the server where the RDBMS is installed. Optionally, specify the server's port number and the database name.



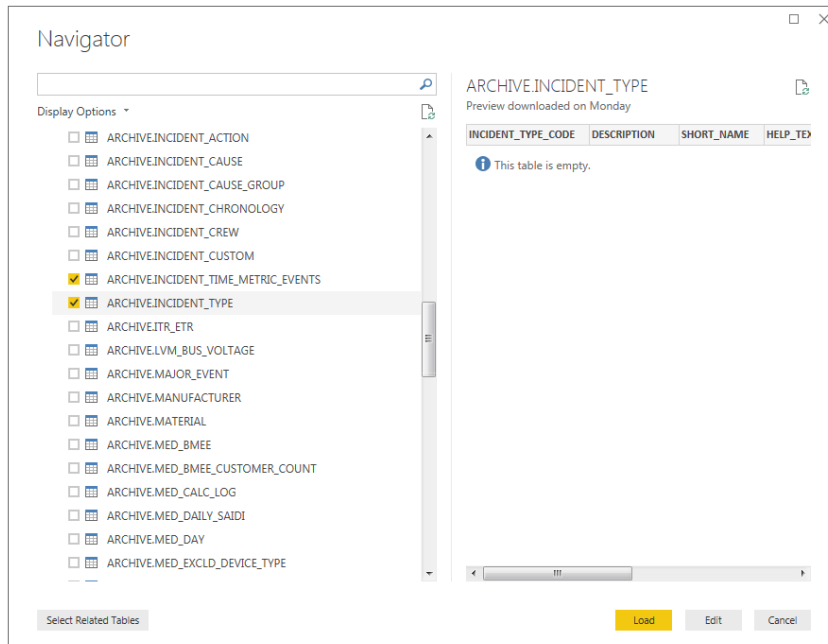
5. Enter the ARCHIVE database user name and password and click Connect.



When you connect to a database, the Navigator window appears and shows all of the database tables. Selecting a table shows its data in the right pane.



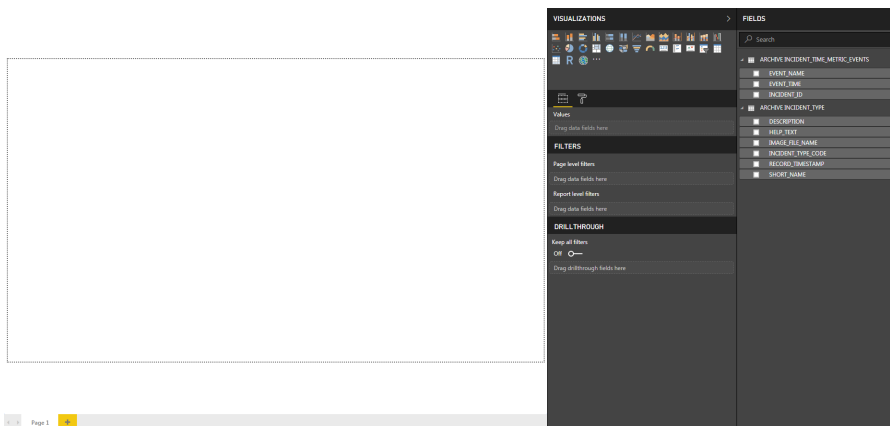
6. In the Navigator window, select the database tables to be used for a report.
7. Click Edit to customize or alter fields, add new columns, or remove existing columns before loading them into a report.
8. Click Load.



Note

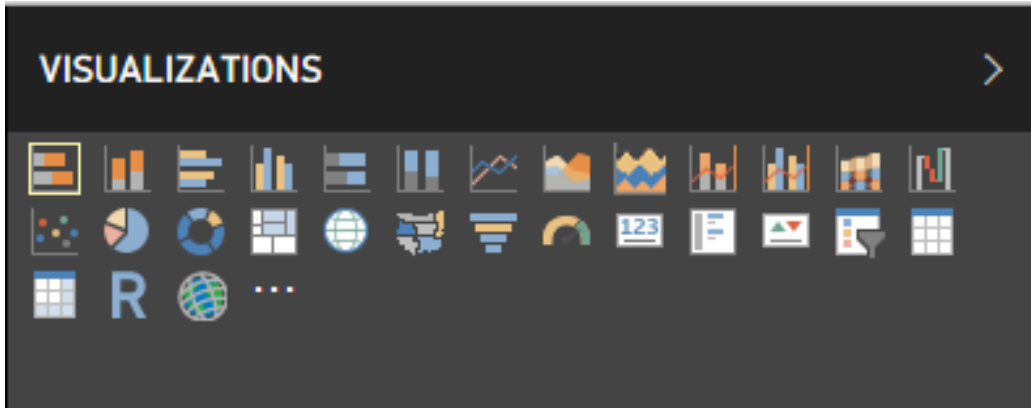
When selecting multiple tables, there must be a relation between them. Otherwise, the report will throw an error when a column or field from a table with no relation is selected.

Once data is loaded, the Report Developer view appears.



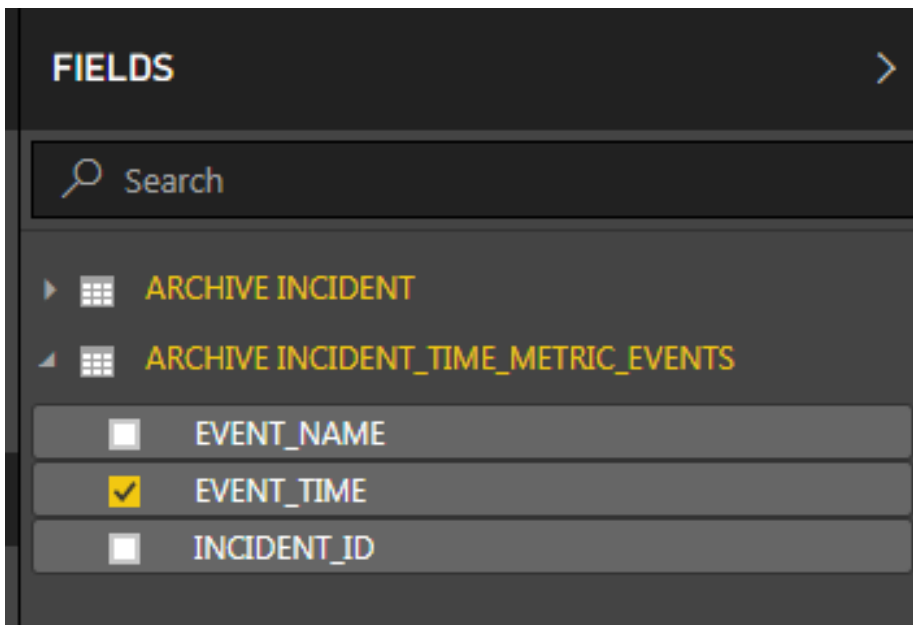
The Report Developer view includes the following sections:

- **Report view:** The blank white area in the initial screen. This area is populated when a report type is selected in the Visualizations section.
- **Visualizations section:** Includes options for choosing the type of report to generate, such as a table or chart.

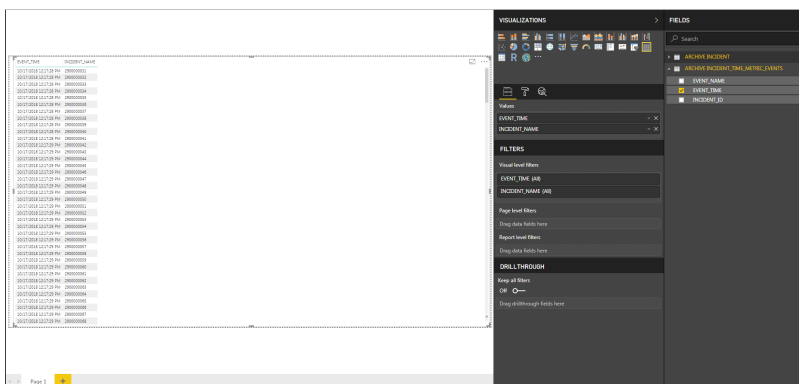


The Visualizations section also includes options to filter and format fields.

- **Fields section:** Includes columns of the tables that are selected (loaded) for the report.



For example, the created Table report looks as follows:



9. Click Save to save the report on your local computer.

You can later edit or publish saved reports.

For more details about creating and publishing reports, refer to Microsoft's Power BI help.